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Centers for Medicare & Medicaid Services (CMS)
Assistant Secretary for Technology Policy (ASTP)/
Office of the National Coordinator for Health Information Technology (ONC)

Request for Information (RFI)
Health Technology Ecosystem

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ACRONYM TABLE

Acronym	Definition
ACO	Accountable Care Organization
API	Application Programming Interface
ASTP	Assistant Secretary for Technology Policy
CDS	Clinical Decision Support
CMS	Centers for Medicare & Medicaid Services
CSP	Cloud Service Provider
EHI	Electronic Health Information
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
HHS	U.S. Department of Health and Human Services
HIE	Health Information Exchange
IAL2	Identity Assurance Level 2
IT	Information Technology
MA	Medicare Advantage
MIPS	Merit-based Incentive Payment System
NIST	National Institute of Standards and Technology
NPI	National Provider Identifier
ONC	Office of the National Coordinator for Health Information Technology
PBM	Pharmacy Benefit Manager
QHIN	Qualified Health Information Network
REACH	Realizing Equity, Access, and Community Health
SDOH	Social Determinants of Health
TEFCA	Trusted Exchange Framework and Common Agreement
USCDI	United States Core Data for Interoperability

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1.0 OVERVIEW

CORPORATE PROFILE

Founded in 2013, Precise Software Solutions, Inc. is an 8(a) Small Business specializing in agile, innovative solutions across mission critical digital transformation, system modernization, cloud implementation, and data analytics initiatives. Our commitment to quality is reflected in our CMMI Level 3 for Development appraisal and ISO 9001, 20000, and 27000 certifications. Precise has supported complex IT modernization initiatives for a broad range of federal agencies, including **FDA, CDC, NIH, HHS (ASPR and ACL), NOAA, NIST, DOL, IRS, and DEA**, as well as **12 states and over 50 local health organizations**. Our performance is consistently rated "Exceptional" in CPARs, particularly in program management, quality, schedule, and cost control. Precise is recognized for its commitment to innovation. Since 2019, we have earned nine innovation awards and one patent, with several award-winning solutions applied directly to FDA programs. Recent industry awards include the 2024 FORUM Innovation Award, 2023 FedHealthIT Disruptive Tech Program Award, 2023 ACT-IAC Innovation Impact Award, 2022 GSA Applied AI Challenge Finalist, 2022 FedHealthIT100 Award Winner, 2022 / 2021 / 2019 Government Innovation Award, 2021 FDA Smart Food Traceability Winner, 2020 / 2019 ATC-IAC Igniting Innovation Top 82021 & 2022 Government Innovation Awards, and the FDA Smart Food Traceability Challenge.

As a **Microsoft Azure reseller** and **Appian Advantage Services Partner**, we deliver **low-code and cloud-native solutions** across **AWS, Azure, and GCP**. We hold **AWS Advanced Tier Partner** status and are progressing toward the **AWS Migration Competency**, ensuring secure, scalable, and compliant implementations across **on-prem, hybrid, and multi-cloud environments**.

INTRODUCTION & EXECUTIVE SUMMARY

Precise Software Solutions, Inc. (Precise) appreciates the opportunity to provide input in response to CMS's Request for Information (RFI) regarding the development of a scalable, secure, and inclusive digital health technology ecosystem. Our recommendations reflect extensive experience supporting CMS, FDA, and broader HHS health IT initiatives with a focus on cloud modernization, API adoption, and regulatory interoperability.

We recognize the significant groundwork laid by the 21st Century Cures Act, ONC Health IT Certification Program, and CMS interoperability mandates. However, persistent challenges—including fragmented infrastructure, voluntary participation, uneven API adoption, and limited data exchange incentives—continue to hinder widespread network scalability and value realization. True interoperability demands not only standardized technical frameworks, but also aligned incentives, shared governance, and sustainable infrastructure investment.

To address these gaps, our response outlines key strategies for building a future-ready digital health ecosystem. These include:

- **Scaling Equitable Data Exchange**: Standardize bulk FHIR API usage across CMS programs and mandate open data exchange to unlock population-level insights, improve public health responsiveness, and reduce administrative burdens.
- **Strengthening Incentives & Participation**: Tie TEFCA-aligned network participation to CMS value-based care models and performance programs (e.g., ACO REACH, MA Stars, MIPS) to ensure active ecosystem engagement.
- **Enabling Infrastructure & Governance Support**: Promote a national FHIR endpoint directory, modern consent frameworks, and identity services to reduce integration friction and increase trust across networks.
- **Expanding API Coverage & Interoperability Standards**: Extend access beyond USCDI to full EHI, including imaging metadata, clinical notes, and SDOH, with consistent implementation of SMART on FHIR and OAuth 2.0 protocols.

- **Promoting Public-Private Ecosystem Development**: Recognize and endorse non-CMS networks that align with federal standards to diversify data sources, support innovation, and extend digital services to underserved populations.

Through coordinated policy leadership and strategic investment, CMS and ONC can accelerate the shift toward a connected, patient-centered data ecosystem—one that delivers measurable impact for beneficiaries, providers, researchers, and public health leaders. We are available to support any follow-up inquiries from CMS.

2.0 TECHNOLOGY VENDORS, DATA PROVIDERS, AND NETWORKS

ECOSYSTEM

TD 1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

To stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers, CMS should pursue a two-pronged approach that combines short-term and long-term actions to build an ecosystem.

In the short term, CMS can significantly lower the barrier to entry for developers by expanding the scope and accessibility of existing infrastructure. Enhancements to the **Blue Button 2.0 API**—such as incorporating additional claims data from **Medicare Advantage (Part C), durable medical equipment (DME), lab results,** and **social determinants of health**—would immediately improve the relevance and utility of CMS data for app development. CMS should also invest in improving the developer experience through a centralized developer portal that offers sandboxes, code samples, and testing tools, along with clear technical documentation and a certification or validation pathway for third-party apps.

To further incentivize participation, CMS could launch time-limited **accelerator programs** targeted at Medicare-specific problems, such as chronic disease management, medication adherence, or caregiver support. These challenges should be designed to support early-stage and small-to-midsize developers, offering funding and technical guidance through proof-of-concept, pilot testing, and implementation. Additionally, CMS could recognize providers and payers that deploy these solutions through bonus points in ACO or value-based purchasing programs, thereby creating downstream demand for new tools.

Long-term, CMS should establish the foundation for a sustainable digital health ecosystem focused on Medicare beneficiaries. This includes building a trusted, CMS-endorsed digital health marketplace where beneficiaries and caregivers can easily find, compare, and authorize applications that integrate with CMS APIs. Such a platform should emphasize transparency, offering user ratings, privacy and security indicators, and—where possible—evidence of clinical or financial impact.

CMS can encourage alignment with national standards and identity frameworks by promoting integration with **TEFCA** and supporting federated digital identity solutions like **Login.gov**. This would reduce friction for both patients and developers, standardizing access across providers and digital services. CMS should also explore partnerships with developers through a formal **CMS Developer Partner Program**, which could offer early access to test data, regulatory guidance, and participation in advisory councils to ensure that future policy aligns with real-world developer needs.

By reducing technical barriers, creating financial incentives, and fostering a culture of open innovation, CMS can continue to transform the Medicare landscape into a vibrant environment where developers are motivated and empowered to create accessible, impactful tools for beneficiaries and caregivers alike.

TD 2 b. Regarding CMS Data, to stimulate developer interest— What additional data would be most valuable if made available through CMS APIs?

Unlocking additional CMS data through secure, well-documented APIs would significantly accelerate innovation across healthcare technology, patient engagement tools, and research platforms. Developers are increasingly seeking real-time, machine-readable access to trusted public health data to power

applications that improve care coordination, reduce costs, and personalize patient experiences. By prioritizing data sets that address cost transparency, social determinants of health (SDOH), quality outcomes, and patient-level insights, CMS can catalyze the next generation of tools that improve healthcare access and equity. In particular, APIs that combine administrative claims data, pricing information, provider performance, and beneficiary experience can serve as the foundation for solutions that empower consumers, inform policy, and drive smarter care delivery models.

Exhibit 1. High-Value CMS Data for Developer Access via API

Category	Description
Cost Transparency	Negotiated prices for Medicare Advantage and Part D, along with out-of-pocket cost estimators and historical cost trend data, enable consumer-facing applications that support price comparison, benefit optimization, and budgeting.
Utilization and Claims Data	De-identified Fee-for-Service (FFS) claims, Medicare Advantage (MA) encounter data, and DMEPOS usage data support analytics on care patterns, development of fraud detection tools, and predictive care applications.
Quality & Outcomes	Hospital Compare data, MIPS/APM performance metrics, and HEDIS and STAR ratings support value-based care dashboards, quality improvement tools, and performance benchmarking.
Provider Directory & Networks	Real-time updates to NPPES, Medicare Advantage provider networks, and PCP assignment APIs support provider finder applications, network adequacy assessments, and referral management solutions.
Beneficiary Access APIs	Blue Button 2.0 extensions—such as integration with Medicaid or social services data—enable developers to create comprehensive patient tools that span the full continuum of care.
Social Determinants of Health (SDOH)	Utilization of Z-codes, SDOH screeners, and area deprivation index overlays supports the development of risk stratification tools and care coordination models that address social and non-clinical needs.
Waivers and Innovation Models	Data on CMMI model outcomes, state waivers, and HCBS services enables developers to create tools that support care innovation and track Medicaid transformation efforts.
Coverage & Enrollment Data	Real-time plan finder data, premium and benefit data APIs, eligibility information drives personalized plan recommendation tools and enrollment assistance platforms.

TD 2 c. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

While Blue Button 2.0 offers a powerful foundation for patient-level insights into Medicare claims data, its value multiplies when integrated with additional data sources that provide clinical nuance, social context, and longitudinal health information. Developers, researchers, and care providers can deliver more intelligent tools and services by combining Blue Button data with electronic health record (EHR) data, social determinants of health, medication adherence insights, and real-time care coordination feeds. These complementary sources transform claims data from a historical record into a predictive, actionable asset that supports preventative care, health equity, and optimized treatment pathways.

Exhibit 2. Most Valuable Data Sources Alongside Blue Button 2.0

Data Source	Description
Electronic Health Records (EHR) via FHIR APIs	• Supplements claims data with clinical observations, labs, vitals, and care plans • Enables real-time medication reconciliation, condition monitoring, and care coordination
Social Determinants of Health (SDOH)	• Provides context on housing, food insecurity, transportation, and socioeconomic status • Supports risk scoring, tailored care plans, and community resource referrals
Pharmacy Benefit Manager (PBM) and Medication Adherence Data	• Offers insight into actual medication fills, adherence patterns, and gaps • Enhances chronic disease management, reduces preventable hospitalizations
Wearables & Remote Monitoring Data	• Brings in biometric data (e.g., heart rate, activity, sleep) • Enables wellness tracking, early detection, and personalized coaching

Data Source	Description
Behavioral Health & Mental Health Claims or Screeners	• Addresses key factors driving cost and outcomes, often underrepresented in general claims • Supports integration of behavioral health in care management platforms
Provider Directory and Network Data (e.g., NPPES, MA Networks)	• Identifies in-network, high-performing, or culturally competent providers • Improves access, referral accuracy, and continuity of care
Price Transparency & Cost Estimators	• Supports financial planning and plan optimization tools • Combines historical Blue Button costs with forward-looking estimates
Health Information Exchange (HIE) Data	• Offers broader view of encounters across care settings • Improves transitions of care and fills gaps missed by Medicare claims alone
CMS Innovation Models & Risk Adjustment Data	• Adds data from ACOs, PCMHs, and CMMI initiatives • Supports advanced population health analytics and value-based program alignment

TD 2 d. What obstacles prevent accessing these data sources today?

While Blue Button 2.0 has improved access to Medicare Fee-for-Service (FFS) claims, key datasets—such as Medicare Advantage (Part C) encounters, prior authorizations, and prescription drug events—remain largely inaccessible due to fragmented data ownership and lack of policy authority. These gaps limit developer innovation and hinder a complete view of patient health.

Many valuable data sources are siloed within private plans, PBMs, or providers, beyond CMS's direct control. Even when technically available, these datasets often lack standardized, API-first infrastructure, especially for lab results, imaging, and DME claims still housed in legacy systems.

The current framework also excludes Medicare Advantage and Part D enrollees, creating inconsistent access across Medicare programs. Additional barriers include the absence of real-time, standardized consent management for sensitive data like SDOH and behavioral health, and general industry reluctance to participate due to risk concerns and limited incentives. Policy fragmentation, outdated systems, and misaligned incentives continue to restrict meaningful data access. Addressing these issues will require coordinated leadership, infrastructure modernization, and clearer participation requirements across the health ecosystem.

DIGITAL IDENTITY

TD 3 a. Regarding digital identity implementation: What are the challenges and benefits?

The implementation of a robust digital identity framework at CMS presents both critical challenges and transformative benefits. As CMS modernizes its infrastructure to support secure, interoperable, and user-friendly digital services, managing digital identity becomes foundational to its success.

Challenges. One of the most pressing challenges lies in achieving interoperability across diverse systems—including legacy platforms, cloud environments, and third-party partners. CMS must align identity solutions across these disparate systems while maintaining compliance with federal standards such as FISMA, HIPAA, and NIST 800-63. Ensuring secure identity proofing and authentication, especially for populations with limited access to digital tools or government-issued credentials, further complicates implementation. Additionally, as CMS serves millions of users—from beneficiaries and providers to contractors and internal staff—scalable role-based access and identity lifecycle management are essential but difficult to implement without creating friction or risk. The transition to a Zero Trust Architecture, which depends heavily on continuous identity verification, also requires substantial investment in automation, governance, and change management.

Benefits. The benefits of digital identity modernization are substantial. A centralized, standards-based identity approach enhances CMS' security posture by reducing the risk of credential misuse and unauthorized access. By enabling capabilities like single sign-on (SSO), multi-factor authentication (MFA), and federated identity, CMS can significantly improve the user experience across Medicare, Medicaid, and

provider-facing systems. These capabilities also support the agency's broader Zero Trust and ICAM goals by allowing continuous access verification and improving visibility across systems. Additionally, centralized identity governance enhances auditability, streamlines compliance reporting, and reduces administrative overhead through automated provisioning and de-provisioning.

Digital identity is a strategic enabler for CMS, allowing the agency to securely scale its services, enhance trust with users, and meet evolving cybersecurity and policy demands across the healthcare ecosystem.

TD 3 b. Regarding digital identity implementation: How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Digital identity credentials through NIST 800-63-3 IAL2/AAL2-compliant providers such as Login.gov or ID.me would help to advance both CMS cybersecurity and data exchange. From a security standpoint, it raises the bar for identity assurance, enabling CMS to better protect sensitive data and align with Zero Trust principles. From a data exchange perspective, these verified identities enable more seamless, secure, and privacy-preserving interactions across the healthcare ecosystem. This approach not only bolsters CMS' ability to safeguard public trust but also empowers beneficiaries, providers, and partners with secure, streamlined access to the services and data they need.

Strengthened User Authentication. Implementing identity verification at IAL2 and requiring AAL2 authentication (e.g., multi-factor authentication) greatly reduces the risk of unauthorized access, account takeovers, and phishing attacks. These CSPs verify identity through government-issued ID and biometric or knowledge-based verification, making identity spoofing more difficult.

Support for Zero Trust Security. Digital identity credentials form a cornerstone of Zero Trust by enabling continuous, risk-adaptive access validation and helping enforce least-privilege access policies across systems and users.

Improved Incident Response and Forensics. Centralized identity providers generate high-fidelity logs that improve traceability, auditing, and incident triage. This enables CMS to detect anomalies and respond to threats more rapidly.

Reduced Credential Management Risk. By federating authentication through trusted CSPs, CMS reduces the burden and security risks associated with managing passwords and local credential storage across multiple systems.

Enhanced Trust Across Ecosystems. Credentialing users through IAL2/AAL2-compliant providers creates a verifiable trust anchor for secure data exchange across CMS systems, providers, states, and third parties. This fosters interoperability between systems (e.g., Blue Button 2.0, Health Information Exchanges) by ensuring all actors meet the same digital identity assurance requirements.

Improved Data Privacy and Consent Management. With strong identity proofing, users can be more confidently matched to their health records and can securely grant or revoke consent for data sharing, a key enabler for patient-directed exchange.

Streamlined Onboarding and Cross-Agency Access. Federated digital identities reduce onboarding friction for users (e.g., healthcare providers, researchers, contractors) by allowing them to access multiple services using a single, trusted credential.

Improved Compliance Data Protection Compliance. Verified identities simplify compliance with privacy and security mandates under HIPAA, FISMA, and the 21st Century Cures Act by ensuring that only authorized users access sensitive data.

TD 3 c. Regarding digital identity implementation: What impact would mandatory use of the OpenID Connect identity protocol have?

Mandating OpenID Connect (OIDC) across the CMS ecosystem would modernize identity management, strengthen security, and streamline user and system access. As a standardized protocol built on OAuth

2.0, OIDC enables token-based authentication, multi-factor support, identity federation, and real-time access validation—key components of Zero Trust.

Adopting OIDC would simplify integration for developers, improve interoperability with external identity providers, and support secure, role-based access across CMS platforms. It would also enable single sign-on and consistent user experiences across applications like Blue Button, HETS, and Medicaid services.

While legacy systems and smaller partners may need support to migrate, the long-term benefits—enhanced auditability, reduced integration burden, and improved scalability—make OIDC a strategic step toward CMS's identity modernization and interoperability goals.

TECHNICAL STANDARDS AND CERTIFICATION

TD 4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

Promoting open, standards-based APIs enables CMS to advance the goals of the 21st Century Cures Act and ONC interoperability rules, while also supporting innovation, equity, and competition in digital health. Proprietary APIs often limit data portability, increase vendor lock-in, and obstruct seamless integration between systems. By prioritizing open standards like HL7 FHIR, SMART on FHIR, OAuth 2.0, and OpenID Connect, CMS can create an ecosystem where healthcare data flows securely and consistently across platforms and stakeholders.

To drive this transition, CMS can leverage its roles as both a regulator and a payer. Mandating open APIs in federal contracts, tying program participation to standards compliance, and setting clear deadlines to retire proprietary interfaces are critical policy levers. Providing developers with robust tools—such as sandboxes, test data, and reference implementations—can ease adoption and foster real-world innovation.

Incentives. CMS can encourage provider and plan participation by linking open API usage to quality ratings or reimbursement models. Offering certification or validation tools for API conformance can further build trust and reduce ambiguity across the ecosystem.

Finally, establishing a strong governance framework—one that coordinates versioning, security policies, and stakeholder input—will ensure consistency and sustainability. Partnering with national standards bodies and aligning with Medicaid and state-level initiatives will help minimize fragmentation.

By reinforcing open API adoption through aligned policy, procurement, technical support, and governance, CMS can shape a more interoperable, future-ready healthcare infrastructure—one that supports better data access, lowers systemic costs, and enables more equitable, patient-centered care.

TD 5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide FHIR endpoint directory would be a critical step toward improving interoperability and simplifying health data exchange across the U.S. healthcare system. One of the greatest current barriers is simply knowing where and how to connect—especially as patients engage with multiple providers and systems. To ensure trust and integrity, the directory should be:

- **Federally maintained**—ideally by ONC, potentially in partnership with a QHIN or other implementation body
- **Publicly accessible**, with optional premium services (e.g., real-time monitoring) offered by private partners
- Designed to promote data transparency, equity, and usability across all stakeholders

A national FHIR endpoint directory would eliminate a fundamental barrier to digital health innovation and empower patients, providers, and payers to connect and exchange health information efficiently, equitably, and securely.

TD 6. What unique interoperability functions does TEFCA perform?

The TEFCA performs several unique interoperability functions that position it as a foundational element of nationwide health information exchange. While many existing frameworks and networks support data exchange regionally or within specific networks (e.g., HIEs, EHR vendor networks), TEFCA introduces standardized governance, legal, and technical mechanisms to unify these efforts under a single, nationwide framework. TEFCA plays a uniquely integrative role in the U.S. health IT ecosystem by establishing a federated, interoperable environment for health data exchange across disparate networks, technologies, and organizations. Developed by the Office of the National Coordinator for Health IT (ONC), TEFCA formalizes the "rules of the road" for nationwide trusted data exchange, enabling participants to access, query, and share electronic health information (EHI) at scale—regardless of vendor, geography, or network affiliation. Unlike traditional HIEs or bilateral exchange agreements, TEFCA provides legal, technical, and governance alignment across a broad range of stakeholders, including payers, providers, public health agencies, and digital health developers.

TD 6 a. What existing alternatives should be considered?

Before fully adopting TEFCA, healthcare organizations, payers, and digital health platforms may consider existing interoperability solutions such as Carequality, CommonWell, and FHIR-based APIs. These alternatives, while often limited to specific networks or use cases, offer mature and widely adopted means of health data exchange that may fulfill short-term objectives.

However, TEFCA is not intended to replace these systems, it is designed to federate them under a common legal, technical, and governance framework. TEFCA introduces a scalable, policy-aligned approach to nationwide health information exchange through designated QHINs, many of which may include or interoperate with existing networks. As such, organizations should view Carequality, CommonWell, and FHIR APIs not as alternatives to TEFCA, but as potential building blocks within the TEFCA ecosystem. A phased approach that aligns immediate interoperability needs with TEFCA readiness can maximize value while ensuring long-term compliance and scalability.

Exhibit 3. Key Alternatives to Consider

Alternative	Description
Carequality	Nationwide data sharing framework connecting EHRs, health systems, and vendors
CommonWell Health Alliance	Vendor-neutral health data exchange network with master patient index and record locator services
eHealth Exchange	Nationwide network supporting federal agencies and healthcare systems
State and Regional HIEs	Local or regional networks facilitating provider and public health exchange
FHIR-based APIs (e.g., SMART on FHIR)	Real-time access to structured clinical data via EHR-integrated apps
DirectTrust / Direct Messaging	Secure, point-to-point messaging for clinical document exchange
Health Gorilla, Particle Health, MEDITECH Traverse, etc.	Aggregator or HIE-as-a-service platforms offering API-based clinical data access

TD 6 b. Are there redundant standards, protocols or channels or both that should be consolidated?

Yes, the current U.S. health data interoperability landscape is fragmented by overlapping standards, protocols, and networks—driving up complexity, cost, and inefficiency. Despite progress, the system still relies on legacy infrastructure and competing models, which burden providers, payers, and developers alike. To modernize and streamline interoperability, CMS and its partners should focus on consolidating outdated systems and aligning efforts under unified frameworks such as FHIR and TEFCA. This includes

phasing out older document-based exchange protocols in favor of FHIR APIs, which offer more scalable, modular, and developer-friendly solutions.

Nationwide networks like Carequality, CommonWell, and eHealth Exchange should federate under TEFCA's Qualified Health Information Network (QHIN) model to eliminate redundant governance and infrastructure. Likewise, scattered provider directories and endpoint registries should be unified into a centralized, open-access FHIR directory linked to NPPES or TEFCA structures to improve data routing and reduce administrative friction.

Standardizing identity assurance and consent frameworks—such as using NIST IAL2 and HL7 FHIR Consent Resource—will also support secure, consistent access across systems and jurisdictions.

Consolidating redundant components is critical to unlocking a more efficient, secure, and patient-centered interoperability ecosystem. A strategic shift toward modern standards and shared trust frameworks will accelerate innovation and improve access to health information nationwide.

TD 7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

The USCDI standard defines a minimum set of structured health data classes and elements to be shared for interoperability. For CMS, which oversees some of the nation's largest healthcare programs and data exchanges, USCDI has been instrumental in establishing a common foundation for information exchange across EHRs, health plans, providers, and digital health tools. It has enabled greater consistency in data formats, supported compliance with CMS interoperability rules (e.g., patient access APIs), and advanced the goals of the 21st Century Cures Act.

However, while USCDI sets a critical baseline, it does not yet meet all the nuanced or advanced needs of CMS programs, particularly in areas like value-based care, public health reporting, and whole-person care. The evolving nature of the healthcare ecosystem—especially with growing needs for behavioral health, social determinants of health (SDOH), and longitudinal care data—means that USCDI's scope remains too narrow for some CMS use cases, see Exhibits 4 and 5 for additional USCDI details.

Exhibit 4. Improved Interoperability with USCDI

Area	Impact
Standardization of Core Data	Established a unified language across EHRs and payers for demographics, medications, lab results, and clinical notes, reducing variability and errors in data exchange.
Support for APIs and FHIR Adoption	Anchors the CMS Patient Access Rule by defining the FHIR data payload for claims and encounter data, enabling app developers to deliver patient-facing solutions.
Improved Data Portability	Enables Medicare and Medicaid beneficiaries to access and share a consistent set of clinical data, regardless of provider or plan.
Foundation for TEFCA and Value-Based Care	Serves as a common denominator to support data exchange across QHINs and CMS innovation models.

Exhibit 5. Limitations of USCDI

Limitation	Description
Scope Gaps for CMS Programs	USCDI v1–v3 lack key elements needed for risk adjustment, quality measurement, prior authorization, and SDOH—all of which are essential for CMS's evolving value-based programs.
Delayed Alignment with Business Needs	The pace of USCDI expansion does not always match CMS program timelines, leading to gaps between regulatory requirements and technical feasibility.
Implementation Variability	EHR vendors and payers may interpret or implement data elements differently, leading to inconsistent payloads or missing fields despite USCDI conformance.
Lack of Support for Claims and Administrative Data	USCDI is clinically focused and does not encompass structured claims data formats that are essential for CMS operations and analytics.

USCDI is an enabler of interoperability that has standardized the exchange of core clinical information and supported federal health IT mandates. However, its utility is limited by narrow data scope, inconsistent implementation, and misalignment with CMS-specific data needs such as claims, quality measures, SDOH, and administrative workflows. To fully support CMS's mission in advancing health equity, care coordination, and value-based care, USCDI must evolve faster and more comprehensively, with active input from CMS and its stakeholders.

TD 7 a. Does it contain the full extent of data elements you need?

No, the USCDI (United States Core Data for Interoperability) does not currently contain the full extent of data elements needed from a CMS context. While it provides a standardized foundation for clinical data exchange, CMS programs require a broader and deeper set of data elements to support complex use cases such as claims processing, quality measurement, prior authorization, risk adjustment, social determinants of health (SDOH), and value-based care administration.

USCDI is a critical step toward consistent interoperability, but its focus is largely clinical and encounter-based, emphasizing EHR-sourced information like medications, allergies, immunizations, and lab results. CMS, as both a payer and regulator, needs access to a much richer and more varied dataset that extends beyond direct clinical care documentation to include administrative, financial, behavioral, and social data. Many CMS programs—such as Medicare Advantage, Medicaid Managed Care, ACO REACH, and other CMMI models—depend on granular, longitudinal, and cross-sector data to manage risk, evaluate outcomes, ensure program integrity, and coordinate care. Current USCDI versions (v1–v3) lack many of these critical data elements, making it insufficient as a standalone standard for CMS operational, policy, and oversight needs.

While USCDI is essential for establishing a minimum interoperable dataset, it does not yet reflect the full breadth of data CMS needs to manage complex federal healthcare programs. For CMS to effectively support value-based care, health equity, public health, and cross-program oversight, it requires expanded and more diverse data elements—particularly in the areas of claims, social risk, quality metrics, behavioral health, and LTSS. USCDI expansion is moving in the right direction, but CMS must continue to advocate for faster, more inclusive development of the standard to align with its broad operational and policy missions.

TD 7 b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

The USCDI provides a foundational step toward standardized health data exchange, but it remains insufficient for meeting the complex operational, regulatory, and policy needs of CMS. Designed as a baseline dataset, USCDI includes essential clinical elements—such as demographics, medications, lab results, and clinical notes—but omits many data types critical to CMS's programs. These gaps include claims and billing data, prior authorization details, risk adjustment factors, social determinants of health, behavioral and cognitive assessments, and long-term services data—areas central to value-based care, health equity, public health, and quality improvement.

Even when included, USCDI elements are often implemented inconsistently across EHRs and health IT systems. Data may be incomplete, captured in non-standard formats, or disconnected from the workflows that support CMS operations. This limits the practical utility of USCDI for oversight, analytics, or real-time data exchange. To make USCDI more actionable for CMS, the standard must evolve in both scope and implementation support. Enhancements should include machine-readable specifications (e.g., CSV/JSON), metadata APIs, synthetic test datasets, automated testing portals, and Git-style version tracking with rolling updates and compatibility markers. These improvements would strengthen interoperability, support better validation and adoption, and ensure USCDI data is accessible, consistent, and aligned with CMS's real-world programmatic needs.

TD 7 c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Expanding the USCDI by adding more data elements holds significant promise for advancing CMS's priorities—but it must be approached with caution to avoid scoping challenges that could compromise its effectiveness. For CMS, broader data inclusion would offer tangible value by enabling more precise risk stratification, robust quality measurement, and improved coordination across care settings, especially for high-need populations. Data types such as prior authorization status, structured claims details, social determinants of health, and behavioral health assessments are not just beneficial—they are essential for CMS to deliver value-based, equitable, and accountable care.

However, broadening the USCDI too rapidly or without clear structure could lead to implementation fatigue, particularly among health IT vendors and providers who are already navigating compliance with federal interoperability rules. Without careful prioritization, the standard risks becoming unwieldy, inconsistent, or unevenly adopted. This would undermine its core goal: creating a unified, actionable, and widely implemented framework for data exchange.

To strike the right balance, ONC and CMS should pursue a deliberate, tiered strategy that distinguishes between foundational core data and specialized extensions. Domain-specific models—such as a USCDI+ track for payer-relevant data or public health reporting—would allow CMS to address its unique data needs without overwhelming the broader healthcare ecosystem. A phased rollout, supported by pilot programs and robust implementation guidance, would allow the industry to adapt incrementally while validating the value of new data elements in real-world contexts.

Adding more data to USCDI can and should create value—but only if expansion is governed by clear priorities, stakeholder alignment, and thoughtful execution. For CMS, this presents an opportunity not just to influence the standard, but to help shape a future where interoperability drives measurable improvements in care delivery, accountability, and population health.

TD 7 d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

From a CMS perspective, the trade-off between structure and coverage—especially in light of improvements in language models and AI processing—raises a timely and nuanced question. The answer depends on the use case, the regulatory context, and the criticality of data accuracy. That said, there is a growing case for strategically accepting less-structured formats in specific scenarios, provided they are paired with advanced NLP capabilities and strong governance frameworks.

CMS has favored highly structured data formats to ensure consistency, comparability, and regulatory compliance. Standards like USCDI, HL7 FHIR, and X12 were designed to enable precise, rule-based data exchange across diverse systems and programs. However, this emphasis on structure often comes at the cost of coverage—leaving out critical insights embedded in clinical notes, prior authorization documents, care plans, and social risk assessments that do not map neatly to discrete data fields.

With recent advances in language models and natural language processing (NLP), CMS now has a compelling opportunity to reconsider this trade-off. Modern models can extract relevant information from unstructured or semi-structured text with increasing accuracy and speed. This opens the door to embracing non-proprietary but less-structured formats (e.g., plain text clinical notes, PDF attachments, or narrative prior authorization requests) as valid inputs to analytics, decision support, and program integrity workflows—particularly when structured alternatives are not available or fail to capture contextual nuance.

The potential upside is considerable: greater data coverage, faster onboarding of non-standard data sources, and reduced burden on providers who struggle with structured data entry. This is particularly

relevant in areas where structured standards lag behind real-world practice—such as behavioral health, social determinants of health, or home-based care documentation.

However, this shift is not without risk. Relying on NLP to interpret unstructured data introduces variability in accuracy, and any errors could affect outcomes like payment decisions, quality scores, or beneficiary support. For mission-critical workflows—such as adjudication, appeals, or compliance audits—structured data will likely remain the gold standard. But for supplementary insights, risk screening, or care coordination, less-structured formats can offer significant value if paired with transparent model validation, audit trails, and human oversight.

In short, yes—CMS can and should selectively accept less-structured, non-proprietary formats, particularly as language models continue to mature. The key is to treat them as complementary to, not replacements for, structured data, and to ensure they are integrated through a lens of governance, validation, and policy alignment. By doing so, CMS can unlock broader data coverage without compromising trust, equity, or accountability—turning modern AI capabilities into a powerful enabler of smarter, more inclusive healthcare delivery.

TD 8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

Under the ONC Health IT Certification Program, the most effective certification criteria and standards are those that directly advance interoperability, patient access, and standardized data exchange—particularly in alignment with CMS and ONC regulatory priorities under the 21st Century Cures Act. These criteria are foundational for enabling nationwide health information exchange, supporting APIs, and ensuring clinical data is shared in a consistent, secure, and usable format.

The ONC Health IT Certification Program defines a set of technical and functional criteria that health IT modules must meet to be considered certified. Among these, a subset has emerged as especially effective in driving interoperability, particularly for CMS-related programs. These criteria support patient data access, FHIR-based APIs, structured clinical content, and standardized vocabulary usage, all of which are central to the federal vision of a connected, patient-centered health system.

The most effective ONC Health IT certification criteria are those that mandate FHIR APIs, enforce EHI export, enable quality measurement, and secure health data exchange—all aligned with national policy goals and CMS program requirements. These criteria not only support compliance but also build the technical foundation for a healthcare ecosystem that is open, secure, and interoperable by design. Moving forward, CMS and ONC should continue to emphasize these standards while driving maturity and adoption across all certified systems.

TD 9 a. Regarding certification of health IT: What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Redefining health IT certification to prioritize API-enabled capabilities over traditional software functionality marks a critical shift toward real-time interoperability, patient access, and system innovation. Traditional certification focused on internal features like recording vitals or generating summaries, but today's healthcare environment requires data to flow seamlessly across systems.

An API-centered certification model—focused on standards like FHIR and SMART on FHIR—enables secure data sharing, supports third-party apps, and fosters a more open, modular ecosystem. This approach empowers developers to build targeted tools such as remote monitoring apps or SDOH dashboards without replicating legacy EHR features, accelerating innovation and reducing vendor lock-in.

For CMS, this evolution supports scalable programs like the Patient Access API and Prior Authorization API, which rely on timely, standardized data exchange. To succeed, certification must also include strong conformance testing, security safeguards, and modular pathways to ensure safe, reliable, and equitable

participation. Prioritizing API functionality shifts health IT from siloed systems to dynamic platforms—advancing CMS's goals for a connected, patient-centered, and efficient healthcare system.

TD 9 b. What would be the drawbacks?

Redefining health IT certification to prioritize API-enabled capabilities holds clear benefits, but it also introduces significant challenges that must be addressed. While APIs offer flexibility and support modern data exchange, shifting certification away from traditional software functionality could impose technical and financial burdens on smaller or under-resourced providers who lack the infrastructure to meet new requirements. There are also risks of inconsistent implementation—APIs may expose data, but without standardization and validation, that data may be incomplete or unusable in real-world workflows. Security concerns rise as APIs increase the number of access points to sensitive health information, especially if not paired with strong privacy and access controls. Additionally, focusing on backend connectivity may neglect usability, leaving APIs poorly integrated into clinical workflows and reducing their value.

In summary, prioritizing APIs in certification is a step toward interoperability, but without thoughtful execution, it could widen disparities, reduce data quality, and compromise security. CMS and ONC must pair this shift with robust governance, implementation support, and a continued focus on usability and equity.

TD 9. c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

ONC and ASTP must evolve health IT certification criteria to require that APIs consistently support access to all aspects of a patient's chart—not just structured data fields, but also free text, scanned documents, faxed records, and other unstructured content. Today, certified APIs focus primarily on standardized, discrete data elements like medications, lab results, and vital signs. While essential, this leaves out large volumes of clinically relevant information that remain locked in narrative notes, attachments, and external documents—often the very materials needed to inform care decisions, referrals, and care coordination.

Redefining certification to include these elements would bring API capabilities in line with how care is documented in the real world. It would ensure that when a provider or patient accesses data through an API, they receive a comprehensive, longitudinal view of the patient's health record, not just fragments limited to structured fields. By requiring APIs to include free-text clinical notes, scanned records, and faxed documents using existing FHIR resources like Document Reference, Binary, and Provenance, ONC can close a critical interoperability gap.

Incorporating these requirements into certification would also support broader policy goals, including TEFCA alignment, patient empowerment under the 21st Century Cures Act, and data completeness for value-based care programs. This shift would need to be supported by rigorous real-world testing, standard metadata tagging, and clear expectations for how unstructured content is made accessible and usable.

Revising certification in this way ensures that APIs are not just technically compliant—they are clinically meaningful. They become tools not only for access, but for understanding the full story of a patient's care, supporting better outcomes, safer decisions, and a more connected healthcare experience.

TD 9. d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

To ensure that API-based data exchange is not only technically possible but also clinically meaningful, CMS must move beyond enabling access and begin holding providers accountable for the completeness and quality of the data they share. While certified health IT systems are now required to support standardized APIs, many providers respond to data requests with only partial or poorly structured records, limiting the utility of these tools for patients, payers, and care teams. The technology exists, but its impact depends on how it is used.

CMS is positioned to address this gap. By aligning policy levers—such as payment models, reporting requirements, and conditions of participation—with expectations for data quality and coverage, CMS can drive real behavior change. For example, participation in value-based care programs like ACO REACH, MIPS, or Medicaid managed care could include requirements or scoring incentives tied to API response completeness, conformance to USCDI standards, or timeliness of data delivery. Likewise, CMS could publicly report on provider performance metrics such as API latency, data coverage, or patient app usage to encourage peer accountability and transparency.

CMS could also define and enforce benchmarks for "data sufficiency" in API responses, ensuring that providers consistently share structured clinical notes, medications, labs, and other essential data—not just a minimal data payload. These benchmarks could be validated through automated reporting or attestation. In parallel, CMS could consider making comprehensive API responses a condition of participation in Medicare and Medicaid, particularly for hospitals, payers, and systems that manage large populations.

To support adoption, CMS might offer technical assistance or enhanced reimbursement to providers that demonstrate high API performance or partner with patients and payers to improve data exchange. Importantly, these policies must be paired with equity safeguards to ensure that smaller or under-resourced providers have the tools and support they need to comply.

In short, if CMS wants APIs to become more than a checkbox—if it wants them to drive better care, coordination, and outcomes—it must create the right policy environment. By tying accountability and incentives to data quality, CMS can ensure that the promise of interoperable health IT translates into real-world value for all stakeholders.

TD 9. e. Regarding certification of health IT: How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

EHRs with bulk data transfer capabilities, particularly those leveraging the FHIR Bulk Data ($export) standard, offer CMS a compelling opportunity to reduce the administrative burden on providers for reporting quality performance data. Today, providers often struggle with fragmented reporting workflows, requiring manual chart reviews, customized queries, and coordination with third-party vendors to meet quality program requirements. This process is time-consuming, costly, and often disconnected from clinical workflows.

By enabling automated, large-scale extraction of standardized clinical data, certified EHRs can streamline how data is pulled from provider systems and submitted to CMS or its authorized intermediaries. Instead of relying on manual submissions or point-to-point interfaces, providers could authorize CMS or a qualified clinical data registry (QCDR) to access relevant patient-level data through secure API-based exports, reducing duplication and freeing up staff to focus on patient care.

To fully realize this benefit, several capabilities must be in place. EHRs must support FHIR Bulk Data APIs that can export full clinical datasets for specific patient cohorts—such as attributed ACO members or Medicare beneficiaries. These data exports must be structured to align with CMS's electronic clinical quality measures (eCQMs) and include robust provenance and metadata to ensure traceability and trust. Additionally, CMS must support feedback mechanisms that allow providers to validate and improve their data prior to final submission.

While the potential is clear, this approach comes with challenges. Many EHRs vary in how consistently they implement FHIR standards, and data completeness—particularly for quality-relevant elements like problem lists, medications, or assessments—remains a concern. Bulk data transfers also raise security and privacy risks due to the volume and sensitivity of the data being exchanged. Furthermore, while clinical burden may decrease, technical demands on IT teams may increase as systems scale up to support these exports reliably.

In summary, EHRs with bulk data transfer capabilities could significantly improve the efficiency, accuracy, and timeliness of quality reporting to CMS. With the right safeguards, standards alignment, and support for data validation, this approach can shift the reporting process from a burdensome task to a seamless, automated component of value-based care, benefiting both providers and the patients they serve.

TD 10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

To fulfill the full intent of the 21st Century Cures Act's API Condition of Certification, ASTP and ONC must go beyond baseline technical compliance and ensure that certified APIs enable real-world, seamless access to the entirety of a patient's electronic health record (EHR)—without special effort. While the current certification framework has successfully mandated standardized FHIR-based APIs, many implementations fall short of the Act's purpose. In practice, some APIs limit access to only a narrow subset of data, omit key unstructured or historical records, or introduce friction that inhibits widespread use by patients, providers, and third-party developers.

The statutory language makes clear that APIs must provide access to all data elements of a patient's EHR, to the extent permitted by law, and that such access must occur without special effort. To bring this vision to life, ONC should take additional steps to ensure completeness, usability, and accountability in certified API implementations.

ONC should clarify that "all data elements" includes not just structured data like vitals and medications, but also clinical notes, scanned documents, test results, patient-reported outcomes, and historical records. Certification should validate that these data types are made accessible through standard FHIR resources—such as DocumentReference, Binary, and Observation component—rather than being siloed within the EHR's interface. ONC must define and enforce what it means to access data "without special effort." This includes prohibiting excessive onboarding requirements, delays, or restrictions on access for patients or authorized apps. APIs should be readily discoverable, reliably functional, and accompanied by robust documentation that supports real-time, secure use. ONC should require developers to demonstrate data completeness and usability in real-world conditions. This could include scenario-based testing, publication of performance metrics (like data coverage or error rates), and active monitoring of API behavior post-certification. Making these metrics publicly available would create transparency and allow stakeholders to assess whether an API is not just certified, but truly usable.

Finally, ONC should continue to coordinate closely with CMS to align incentives—tying participation in federal programs to providers' use of high-performing, standards-conformant APIs. Together, these agencies can help shift the culture from minimal compliance to meaningful, patient-centered data exchange. Realizing the promise of the 21st Century Cures Act will require ONC and ASTP to enforce not just **access**, but **effective access**—ensuring APIs serve as robust, equitable, and comprehensive tools for health information exchange. By strengthening certification to reflect the law's spirit and not just its letter, ONC can advance a health IT ecosystem that truly empowers patients, reduces burden, and fosters innovation.

TD 11. a. Should this capability be revised to specify standardized API requirements for EHI export?

Now that the electronic health information (EHI) export capability has been in production for over a year, it is evident that the lack of standardization in its implementation has limited its impact. While the original intent—to give health IT developers flexibility—was understandable in the early stages, that flexibility has

resulted in widely variable, non-interoperable solutions that often fall short of enabling meaningful data access, especially at scale. As a result, the core goals of the 21st Century Cures Act—to support patient empowerment, reduce administrative burden, and improve data liquidity—are not being fully realized.

To address this, CMS and ASTP/ONC should now revise the EHI export certification requirement to mandate the use of standardized APIs, specifically aligning with the FHIR Bulk Data Access API ($export) for population-level exports and standard FHIR R4 APIs for individual patient access. These APIs are already supported under other certification criteria and have robust implementation guides that promote consistency, scalability, and machine-readability. Standardizing on these APIs would help eliminate the need for custom interfaces or manual workarounds, making EHI export more practical, secure, and usable in real-world settings.

Requiring standardized APIs would also enable downstream benefits. Providers could automate data sharing for quality reporting, population health, or research. Patients could more easily access and transfer their complete records through apps. And third-party developers, researchers, and public health agencies could work with consistent data structures, dramatically reducing integration time and cost.

In short, the time for flexibility has passed. With over a year of real-world experience, CMS and ONC are now in a strong position to enforce consistency, usability, and interoperability. By revising the EHI export requirement to use standardized, widely adopted APIs, they can ensure that the capability delivers on its intended value—turning static compliance into meaningful access.

TD 11. b. Are there specific workflow aspects that could be improved?

Since the EHI export capability became a requirement in January 2024, many health IT developers have technically complied, but the workflows surrounding its use have revealed significant gaps. The flexibility afforded to developers in how they implemented EHI export has led to a wide range of user experiences—many of which are inconsistent, inefficient, and burdensome for patients, providers, and IT staff.

For patients, the process of requesting and receiving their health data is often confusing. In some cases, patients must navigate multiple portals, fill out forms, or wait extended periods for manual responses. Even when data is delivered, it may arrive in formats that are difficult to interpret or use—limiting its value for sharing with other providers or incorporating into personal health apps.

For providers and administrators, the population-level export process is similarly fragmented. Triggering an export may require multiple manual steps, custom configurations, or back-end support from IT teams. There is often little transparency into the status of export jobs, and audit trails are either missing or inaccessible. This adds unnecessary administrative burden and undermines the efficiency gains the export capability was intended to deliver.

To make EHI export meaningful in practice, the workflows supporting it must become more intuitive, automated, and standardized. Patients should be able to initiate and track exports directly from their portals, with clear instructions and access to structured, secure data. Providers should be able to perform bulk exports with minimal configuration, supported by dashboards, export status indicators, and audit logs. Organizations should have clear protocols for who can initiate exports, how consent is managed, and how the resulting data is delivered and used.

Ultimately, the current variability in workflows is a barrier to realizing the full promise of the EHI export requirement. To address this, CMS and ONC should promote workflow standardization and integration, encouraging developers to move beyond basic compliance and focus on user-centered, efficient implementations that support seamless data access for all stakeholders.

TD 11. c. Should CMS consider policy changes to support this capability's use?

Yes, CMS should consider strategic policy changes to actively support the use of the EHI export capability, ensuring it becomes a practical tool for interoperability rather than a passive regulatory requirement. While many health IT developers have technically met the certification standard, the

capability is often underutilized, and workflows to initiate, manage, and apply EHI exports remain fragmented and inconsistent. Without strong policy alignment, the export function risks remaining on the sidelines—available in systems but disconnected from the real-world needs of providers, patients, and data users. To fully realize the intent of the 21st Century Cures Act, CMS can play a pivotal role in moving EHI export from compliance to action by integrating it into its programs, reporting mechanisms, and value-based care initiatives. For example, CMS could incentivize the use of standardized EHI export workflows in accountable care programs like ACO REACH or MIPS, recognizing providers who automate and operationalize data exports for care coordination or population health management. This would shift the capability from being reactive to becoming a proactive tool for performance and equity.

CMS could also align quality reporting and administrative simplification efforts with EHI export functionality. Encouraging the use of FHIR-based bulk data exports to fulfill data submission requirements would reduce reporting burden and enable more consistent, machine-readable data flows. Pairing EHI export with patient access initiatives—such as app-based data sharing or consent-driven exchanges—would empower individuals to take control of their health data and drive innovation in consumer health tools.

To ensure equitable adoption, CMS could provide targeted technical assistance, funding, and implementation guidance for smaller practices or under-resourced providers, helping them incorporate EHI export into routine workflows. Publicly reporting usage metrics or EHI export readiness could also encourage greater transparency and drive vendor accountability.

CMS has an opportunity to elevate EHI export from a static requirement to a foundational enabler of connected, data-driven care. With the right policy levers—focused on incentives, integration, and infrastructure—CMS can ensure this capability delivers real value to the healthcare system and the people it serves.

DATA EXCHANGE

TD 12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

Yes, CMS should consider endorsing non-CMS data sources and networks if they meet clearly defined standards for interoperability, data quality, security, and alignment with CMS's mission. As the healthcare landscape becomes increasingly decentralized, CMS cannot rely solely on its own data systems to support value-based care, population health, and patient-centered innovation. Health information exchanges (HIEs), Qualified Health Information Networks (QHINs) under TEFCA, payer data platforms, registries, and even digital health apps are already aggregating and exchanging valuable health data across the country. By formally recognizing and endorsing trusted external data sources, CMS can promote a more connected and scalable data ecosystem. Endorsement would incentivize private and public sector actors to align with federal interoperability standards, participate in key initiatives like TEFCA, and ensure data is exchanged securely, accurately, and equitably. At the same time, it would expand CMS's ability to access timely, patient-centered data from sources it does not directly control—helping to reduce reporting burden, enhance care coordination, and inform better policy and payment decisions.

However, endorsement should not be automatic. It should be based on clear and measurable criteria to ensure that CMS is promoting only high-quality, standards-aligned partners. These criteria should include technical capabilities (such as FHIR API support and conformance to USCDI), data completeness and accuracy, timely access, privacy and security compliance, and transparent governance structures. Networks should also demonstrate their ability to support CMS-relevant use cases such as quality reporting, public health response, and patient-directed data exchange. Equally important is the ability of

endorsed networks to serve diverse populations, including those in rural and underserved communities, and to incorporate data relevant to health equity, such as social determinants of health.

In short, CMS endorsement of non-CMS data sources is not only appropriate—it's necessary for the agency to stay ahead in a rapidly evolving health data landscape. With the right safeguards and criteria, this approach can strengthen interoperability, promote innovation, and ensure that more stakeholders across the healthcare system are working toward shared goals of data accessibility, accountability, and better health outcomes.

TD 13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

API-based access to full electronic health information (EHI) is a transformative step toward a more connected and patient-centered healthcare system. By moving beyond limited datasets like USCDI, full EHI access brings together structured data, clinical notes, images, care plans, and social determinants of health into a comprehensive, real-time view of a patient's health. This shift turns health data from static and siloed into portable and actionable—fueling smarter care, advanced analytics, and stronger public health capabilities.

With this access, clinicians can leverage AI-driven decision support tools that draw on the full spectrum of patient history to deliver more personalized and proactive care. Patients, in turn, are empowered to control and share their data through apps that support chronic disease management, second opinions, and better care coordination.

Standardized APIs also streamline quality reporting and regulatory compliance by reducing the manual burden on providers and enabling real-time performance tracking. Payers benefit from faster, more efficient prior authorization processes, while public health agencies gain improved tools for outbreak response and chronic disease monitoring through bulk data access.

Researchers can tap into de-identified EHI to support clinical trials, equity-focused studies, and real-world evidence generation, accelerating innovation across the healthcare ecosystem. And during care transitions, providers can retrieve complete, current records to reduce errors and ensure continuity. Full EHI access through APIs is a regulatory milestone and an enabler for delivering smarter, safer, and more equitable care at scale.

TD 13. a. What are the primary obstacles to this?

The opportunity to access the entirety of a patient's electronic health information (EHI) through standardized APIs represents a profound shift in healthcare—offering the potential to transform care delivery, empower patients, advance research, and reduce systemic inefficiencies. Yet despite this promise, several key obstacles continue to impede progress. These challenges are not simply technical in nature, but deeply rooted in the way health data is documented, governed, shared, and utilized.

One of the most pressing barriers is the inconsistency and incompleteness of the data itself. While APIs may be technically capable of retrieving full EHI, the information exposed is often fragmented, poorly structured, or incomplete—particularly when it comes to clinical notes, legacy documents, or data generated outside core EHR modules. This limits the ability of patients, providers, and digital tools to make meaningful use of the information. Compounding this is a lack of standardization across EHI elements beyond USCDI, making data integration and interpretation difficult and resource-intensive for third-party developers and researchers.

Full access to EHI raises valid questions about how sensitive health information—especially behavioral health notes, genomic data, or social determinants—is protected. Current consent models are rarely nuanced enough to accommodate varying patient preferences, and organizations remain cautious about potential risks of data breaches or misuse. Without a clear, scalable framework for patient-managed consent and strong governance across networks, trust in comprehensive data sharing remains fragile.

Another critical obstacle is the limited alignment of incentives. Many providers and health systems receive little direct benefit for ensuring comprehensive, standards-based data availability via API. Without policy levers or payment models that reward transparency and interoperability, the motivation to prioritize full EHI access is often weak—particularly for resource-constrained organizations.

The road to realizing the full value of EHI APIs is blocked not by a single barrier, but by a convergence of technical, policy, and cultural factors. To overcome these challenges, CMS and ONC must promote stronger standardization, enforce the principle of access "without special effort," strengthen privacy and consent frameworks, and create meaningful incentives for data transparency. Only then can the vision of accessible, complete, and actionable health information move from possibility to practice.

TD 13. b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

Accessing full Electronic Health Information (EHI) introduces heightened security and privacy risks, especially when aggregated into a centralized repository or data lake. Full EHI often contains highly sensitive and unstructured information, including free-text clinical notes, behavioral health data, and imaging records—elements that may inadvertently expose personally identifiable information (PII) or protected health information (PHI) not intended for broad use. The larger and more complex the dataset, the greater the attack surface, and the more difficult it becomes to apply consistent data minimization, masking, and access control policies. To mitigate these risks, organizations should implement fine-grained access controls that restrict EHI exposure based on user roles, purpose of use, and patient consent. Additionally, deploying automated redaction and de-identification tools—especially those augmented by natural language processing (NLP)—can help prevent sensitive data from being shared inappropriately while still enabling analytic use. Robust logging, audit trails, and adherence to zero-trust architecture principles further strengthen the security posture when handling full EHI.

TD 14. a. Regarding networks' use of FHIR APIs: How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

Not Applicable.

TD 14. b. Regarding networks' use of FHIR APIs: How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

Connections to FHIR APIs across health data networks are made through a mix of standardized, federated, and proprietary methods—depending on trust levels, technical maturity, and policy alignment. While adoption is increasing under mandates like the ONC Health IT Certification Program and TEFCA, variations persist. Many networks connect through ONC-certified FHIR API endpoints that support patient-level data access using FHIR R4 and SMART on FHIR protocols. These endpoints serve as the foundation for patient-facing apps and are typically integrated via secure OAuth flows.

As TEFCA rolls out, networks and QHINs are beginning to incorporate FHIR alongside legacy standards such as IHE XCA for document exchange. This dual support enables gradual migration to more modern, API-driven workflows, including bulk data access.

Some connections still rely on proprietary APIs that extend beyond USCDI, using vendor-specific authentication or data formats. While they enable advanced features like payer-provider integration, they contribute to interoperability challenges. Many networks use third-party gateways or integration platforms to manage and normalize FHIR traffic across endpoints, enforce policies, and route data through trusted frameworks like Carequality or private HIEs. FHIR connectivity today reflects a hybrid

environment, progressing toward standardization but still shaped by legacy infrastructure and vendor variability.

TD 14. c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

Yes, most modern health data networks using FHIR APIs actively interconnect to expand reach, enable seamless patient data exchange, and improve interoperability. These connections are established through a mix of federal frameworks, national coalitions, and private agreements.

- **TEFCA Integration**. Networks are joining TEFCA through QHINs, gaining standardized legal and technical interoperability with other networks regardless of vendor. TEFCA provides common governance, agreements, and supports future FHIR-based exchange.
- **National Networks (e.g., Carequality, CommonWell)**. Many networks participate in established frameworks offering EHR-agnostic document and FHIR-based exchange. These provide common legal, security, and onboarding protocols.
- **Private Agreements**. Bilateral and regional partnerships are used for specialized needs—such as behavioral health, research, or payer-provider collaboration—often with custom APIs and governance layers.
- **FHIR API Integration**. Standards like OAuth 2.0 and SMART on FHIR enable secure, app-driven access across vendor ecosystems without tightly coupled infrastructure.

These approaches create a growing, interconnected ecosystem where FHIR APIs serve as the technical foundation for scalable, patient-centered data exchange nationwide.

TD 15. a. Regarding bulk FHIR APIs: How would increased use of bulk FHIR improve use cases and data flow?

Increased use of bulk FHIR APIs would significantly enhance data flow and unlock high-impact use cases by enabling efficient, scalable access to large volumes of patient data across systems. Bulk FHIR supports population-level queries, allowing organizations to retrieve data for cohorts—rather than one patient at a time—which is essential for quality measurement, risk stratification, public health surveillance, and value-based care analytics. It reduces reliance on manual data pulls or custom integrations, streamlines reporting, and enables real-time insights across payers, providers, and researchers. Broader adoption would promote faster, more consistent, and cost-effective data exchange across the healthcare ecosystem.

TD 15. b. Regarding bulk FHIR APIs: What are the potential disadvantages of their use?

Bulk FHIR APIs enable powerful population-level data exchange for public health, research, and quality reporting. By allowing extraction of full cohorts, they support deeper insights and more efficient workflows. However, they come with key challenges: high infrastructure demands, heightened security risks, inconsistent vendor support, and complex governance around consent and data use. Bulk data is often difficult to manage and use without advanced filtering and processing. To realize their full value, stakeholders must invest in scalable infrastructure, standardize implementation, and strengthen policies to ensure secure, practical, and equitable use.

TD 16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

As health data exchange evolves, organizations must choose between point-to-point connectivity and shared network infrastructure. Point-to-point offers tailored integration and control, making it effective for bilateral, high-volume exchanges. However, it is resource-intensive, difficult to scale, and prone to fragmentation as each connection requires separate setup and maintenance.

Shared network models—such as TEFCA, QHINs, or FHIR-based HIEs—address these limitations by enabling many-to-many exchange through standardized protocols, centralized governance, and identity frameworks. They improve data liquidity, simplify onboarding, and reduce integration overhead. Still, these models may offer less customization and introduce governance complexities.

Ultimately, point-to-point models offer precision but do not scale well, while shared infrastructure promotes broader, more efficient interoperability. A hybrid approach may persist, but the national trend favors shared models to meet the demands of scalable, secure, and patient-centered data exchange.

TD16 a. Do current rules encourage scalable network participation?

Current federal policies—shaped by the 21st Century Cures Act, ONC Health IT Certification Program, and CMS interoperability rules—have laid essential groundwork for improving health data exchange. These initiatives promote technical readiness, standardized APIs like FHIR, and discourage information blocking. However, they fall short of enabling widespread, scalable participation in nationwide health information networks.

While TEFCA introduces a promising national trust framework, its voluntary nature and lack of financial incentives have limited adoption among providers, payers, and vendors. Many smaller and under-resourced organizations still face high costs, limited technical capacity, and fragmented systems, making meaningful participation difficult.

Enforcement of current rules remains limited, with requirements focused on making data “available” rather than ensuring its effective use. Without stronger incentives or mandates, the operational and financial burden of participation continues to outweigh the perceived benefits for many stakeholders.

To move from intent to impact, CMS and ONC may need to strengthen their approach by linking TEFCA participation to federal program eligibility, integrating incentives with value-based care, and supporting under-resourced providers. Only then can interoperability evolve into a truly scalable, equitable, and unified national system.

TD 16 b. What changes would improve alignment (for example, API unification, reciprocal access)?

To realize scalable and equitable health data exchange, federal rules and infrastructure must evolve to promote stronger alignment across APIs, networks, and stakeholders. Despite regulatory advances, persistent fragmentation—technical, legal, and operational—continues to hinder seamless data flow and erodes trust across the ecosystem. Key improvements include unifying API standards around SMART on FHIR and Bulk FHIR for consistent implementation across vendors and use cases. This would reduce variability and make integration easier for developers, particularly those serving under-resourced providers. Ensuring reciprocal data sharing is also critical. Networks should prohibit one-way data flows and require that all participants contribute data under fair and transparent rules, as outlined in TEFCA and other national frameworks.

A centralized FHIR endpoint directory and standard OAuth2-based trust frameworks would simplify onboarding and interoperability across systems, while national standards for consent and sensitive data handling would strengthen privacy and build confidence in data sharing.

CMS should incentivize participation in TEFCA-aligned networks by linking it to value-based payment programs. Providers who exchange data through certified APIs could receive performance credits or reduced reporting burdens.

Achieving true interoperability requires more than technology—it demands unified policies, shared responsibility, and aligned incentives to ensure health information flows securely and efficiently across the care continuum.

TD 17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

To build a sustainable and impactful health data exchange ecosystem, CMS and ASTP/ONC must take a more coordinated role in shaping both infrastructure and incentives. While federal standards have laid the foundation, long-term growth depends on aligning high-quality data supply with meaningful demand—and ensuring participation is practical and valuable for all stakeholders.

As the largest healthcare payer, CMS can drive engagement by linking data exchange participation to reimbursement programs like ACO REACH, Medicare Advantage Star Ratings, and MIPS. Incentives for using TEFCA-aligned APIs, combined with reduced reporting burdens, can accelerate adoption. CMS should also support small and rural providers with funding and technical assistance to meet interoperability requirements.

ASTP and ONC should continue to enforce consistent FHIR API implementation for bulk data, consent, and full EHI access, while ensuring certified systems support standards-based, vendor-neutral exchange. Investments in shared infrastructure—like a national FHIR endpoint directory and unified identity and consent services—can reduce participation barriers.

To reinforce value, CMS could promote real-world use case pilots, publish data usage metrics, and explore utility-based funding models. With aligned incentives, strong technical standards, and targeted support, CMS and ONC can create a scalable, equitable, and durable nationwide data exchange ecosystem.

3.0 CLOSING

Precise Software Solutions, Inc. appreciates the opportunity to contribute to this important dialogue shaping the future of the CMS health technology ecosystem, We bring deep experience supporting large-scale federal health IT modernization efforts—including secure API development, identity access management, and digital services for regulatory programs. We remain committed to aligning our capabilities with CMS's vision and stand ready to support pilot initiatives, technical implementation efforts, or ecosystem collaboration that advances interoperability, equity, and value-based care.